Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Artificial Intelligence
Intelligent Multi-Agent and Expert Systems (AI401), Semester VII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_10_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 10

Ethical Considerations — Auditing a Decision System for Bias, Privacy and Transparency

Aim

To audit a rule base induced from historical decisions, measuring how its outcomes differ between two groups of applicants, how far the records it was built from can be re-identified, and how much of a decision can be explained to the person it affects.

Theory

Experiment 9 obtained rules from past cases without asking whether those cases were decided well. If the historical decisions treated one group worse than another, the induced rules reproduce that treatment faithfully, and a system that is accurate against its training labels can be systematically unfair. Fairness is measured by comparing outcomes across groups: the selection rate, the proportion of each group accepted; the true positive rate, the proportion of genuinely qualified applicants accepted; and precision, the proportion of accepted applicants who were genuinely qualified. These measures disagree whenever the groups have different base rates, so equalising one generally unbalances another, and the choice of which to equalise is a policy decision rather than a technical one. Deleting the group attribute does not settle it either, because other attributes correlated with it act as proxies.

The same data raises two further questions. Privacy is threatened not by the identifiers that were removed but by the combinations that remain: a set of quasi-identifiers partitions the records into equivalence classes, and a record alone in its class can be re-identified by anyone holding the same attributes from another source. The size of the smallest class, k, is the usual summary, and it is raised by generalising or suppressing attributes. Transparency is the other side of the same structure: an induced rule can be read back to the applicant as the reason for the decision, so the length of the path that produced it bounds how intelligible the explanation is, and the attributes appearing along that path are exactly what the system acted on. Accountability begins with being able to produce that trace at all.

Algorithm

Algorithm 1 — Fairness audit

1. Generate a population of applicants in which a protected group attribute is correlated with the other attributes, record for each applicant both the outcome that is actually deserved and the decision historically taken, and corrupt a fraction of the historical decisions against the protected group. Induce a rule base from the historical decisions, using the method of Experiment 9.
2. On a separate test population, compute for each group the base rate of deserved acceptance, the selection rate, the true positive rate, the false positive rate and the precision of the induced rule base, and report the gap between the groups on each. Report also the agreement of the rule base with the historical decisions and its accuracy against what was actually deserved.

Algorithm 2 — Privacy and transparency audit

1. Choose a set of quasi-identifiers from the same records, group the records into equivalence classes by their values, and report the size of the smallest class, the number of records alone in their class and the proportion lying in classes of fewer than five. Repeat after generalising the quasi-identifiers by merging values and then by suppressing an attribute entirely.
2. For every applicant in the test population, record the sequence of attribute tests the rule base followed to reach its decision. Report the mean and the maximum number of conditions in that sequence, and print the complete trace for one named applicant as the explanation that would be given to them.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the per-group table of base rate, selection rate, true positive rate, false positive rate and precision with the gaps, the table of equivalence class statistics at each level of generalisation, the mean and maximum explanation length, and the full trace for one applicant.

Observations

Comment on how the gap in selection rate compares with the gap in the base rate of deserved acceptance and what that difference means, on which attributes appear in the printed explanation and whether any of them should have been used, on what a record alone in its equivalence class exposes its subject to, and on whether a system that reports its accuracy alone has told its users enough to be held accountable.

Exercises

1. Attempt three mitigations and compare them against the unmodified system: remove the protected attribute from the feature set, remove both the protected attribute and the attribute most correlated with it, and train instead on the outcomes that were actually deserved. Report the accuracy and the gap in selection rate, true positive rate and precision for each, and state which gaps can be closed, which cannot, and why the last configuration is not available in practice.
2. Repeat the induction at each level of generalisation of the quasi-identifiers. Report the smallest equivalence class, the number of unique records, the agreement with the historical decisions, the accuracy against what was deserved and the mean explanation length at every level, and state what generalising the data cost and what it bought.